

MoneyBrew

Phase III: Hi-fi prototype and user evaluation

L.EIC | Human-Computer Interaction

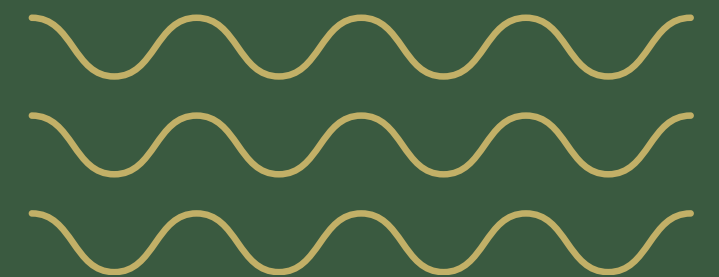
Class 03, group 02

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Project Brief Description

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FINANCIAL TRACKING

MoneyBrew is a **mobile financial management application** whose aim is to bring together all essential aspects of personal finance into a single, secure, and intuitive platform.

The app allows users to automatically track income and expenses through **bank account integration**, visualize their **financial history** using reports and charts, and define savings goals with a **savings plan tracker**. Expense management is supported through categorized spending and the **identification and management of recurring subscriptions**.

GUIDANCE AND EDUCATION

MoneyBrew includes **Brewy**, an AI chatbot that provides finance-related explanations and personalized recommendations, as well as a **financial literacy hub** with articles, videos, and quizzes on topics such as budgeting, taxation, work contracts, and investments.

Smart notifications, personalized tips, location-based recommendations, and price alerts further help users make informed decisions with minimal effort, encouraging healthier financial habits and long-term financial well-being.

Part I and II - Main takeaways

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PART I

USER AND TASK ANALYSIS

Part I: user and task analysis, aiming to understand potential users, their financial behaviors, and their expectations.

Implementation: **questionnaire** with 38 participants, followed by a **PACT analysis**, as well as the creation of personas and activity scenarios.

- Young adults, mainly students or working students, who regularly monitor their expenses and attempt to follow a monthly budget.
- Many participants do not currently use financial management applications.
- Strong interest in features such as automated expense tracking, consumption analysis, financial education, and AI-based support.
- **Main requirements:** simplicity, security and privacy.
- Development of two personas and activity scenarios.

PART II

LO-FI PROTOTYPE AND HEURISTIC EVALUATION

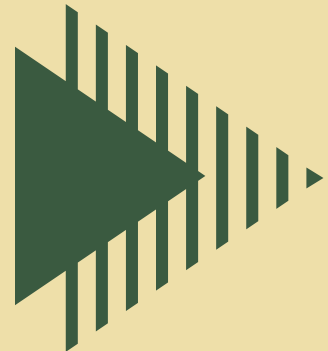
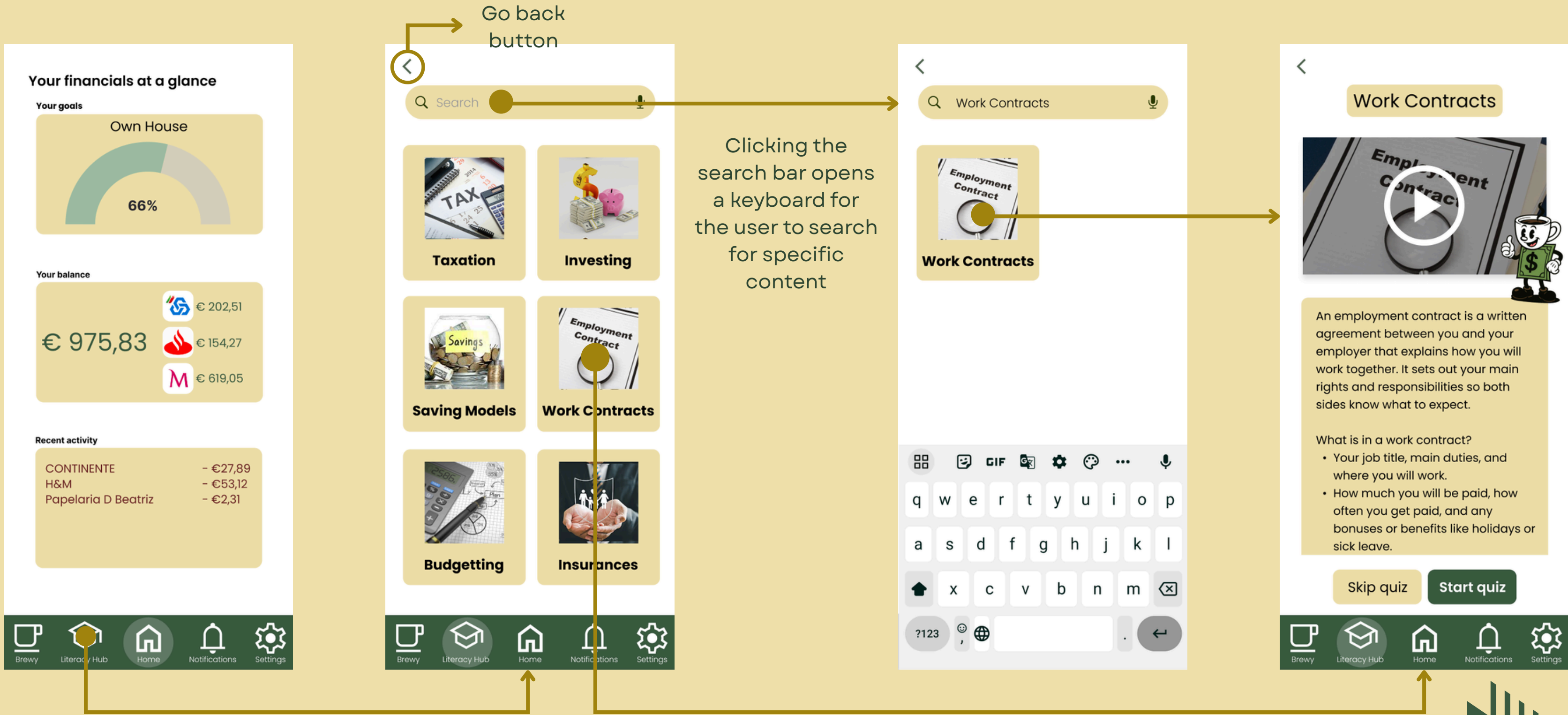
Part II: Lo-fi prototype to explore and evaluate key functionalities of the MoneyBrew app.

Implementation: selection of three **tasks** representing core features of the system: the Brewy AI chatbot, smart and automatic notifications, and a financial literacy hub + **heuristic evaluation** based on Nielsen's usability heuristics.

- The heuristic evaluation revealed several major, minor, and cosmetic usability issues, mainly related to user control and freedom, recognition rather than recall, and visibility of system status.
- **Major issues:** biased decision prompts and limited feedback.
- **Minor issues:** navigation flexibility, lack of progress indicators, and inconsistencies in button labeling.
- Results provided clear guidance for improvements to be implemented in the hi-fi prototype.

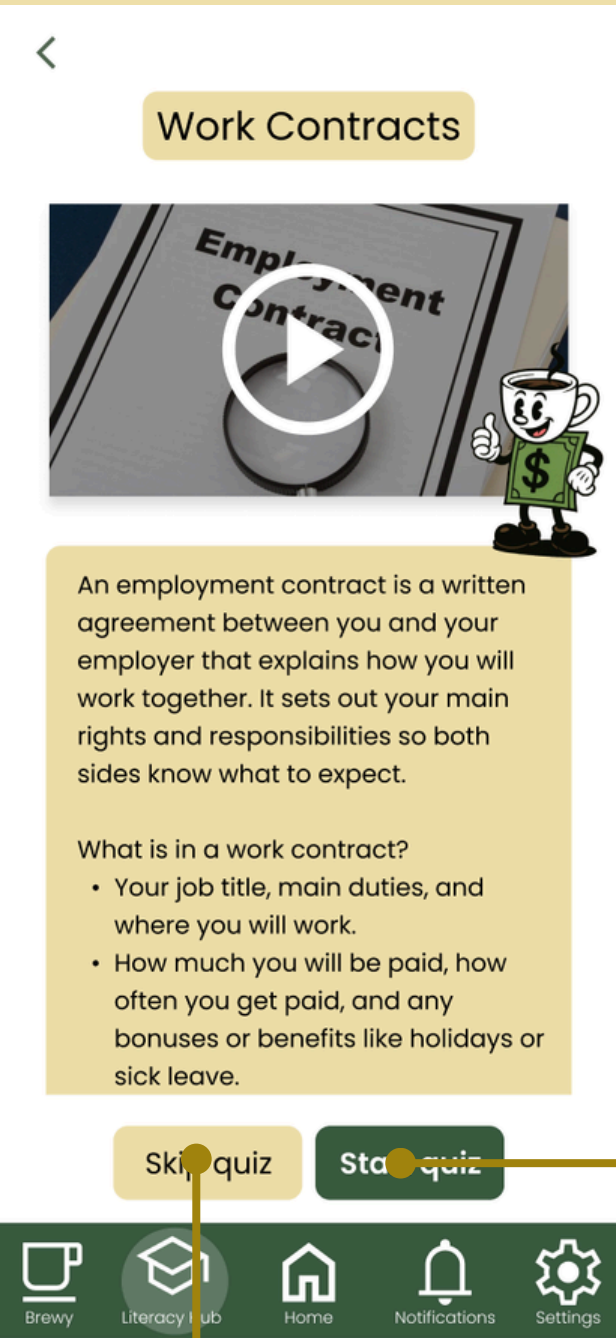
Wireflow - Financial Literacy Hub

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Wireflow - Financial Literacy Hub (cont.)

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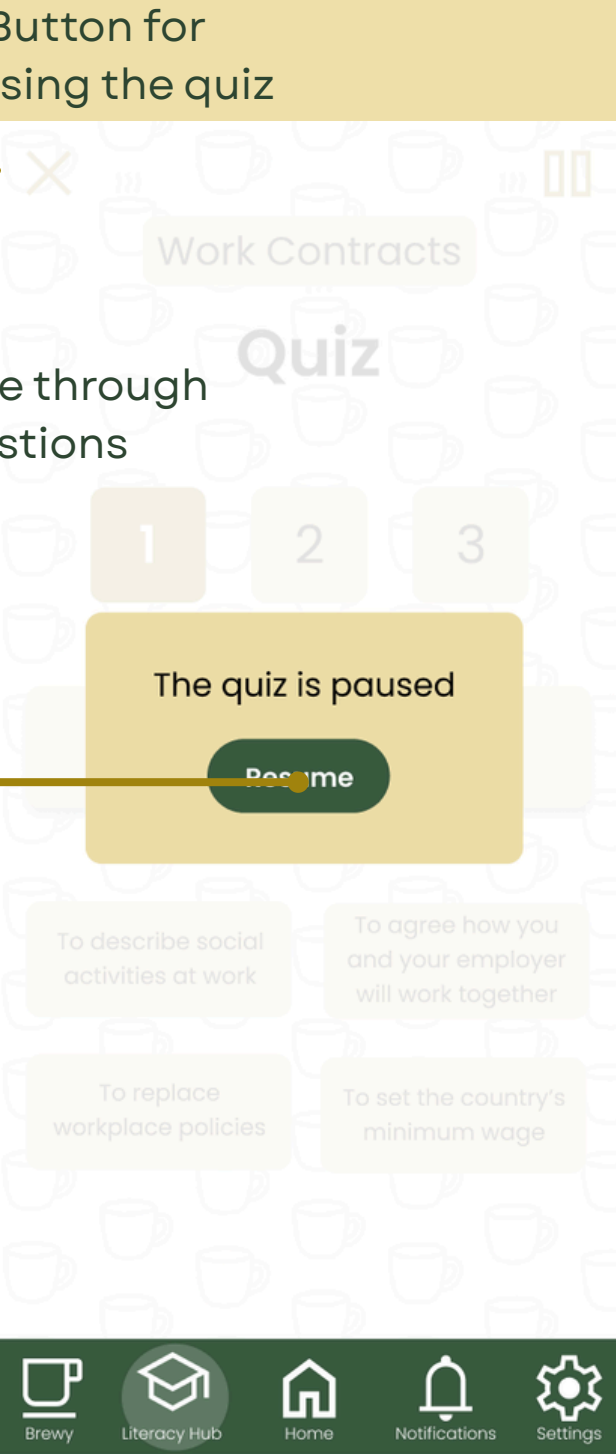


By clicking on the "Skip Quiz" button, the user is redirected back to the literacy hub page

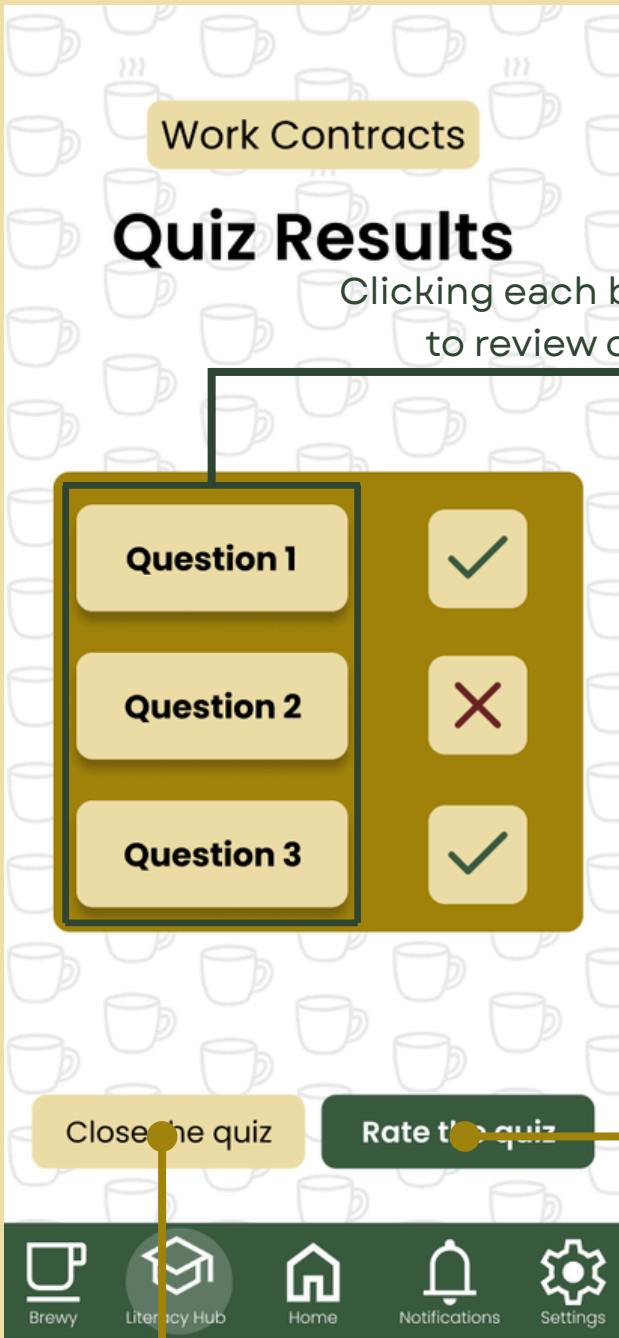
By clicking on the "Start Quiz" button, the user is redirected to the first question



When in a question page, clicking on an option card redirects the user to the next question. If it is the last question, the user is redirected to the feedback page



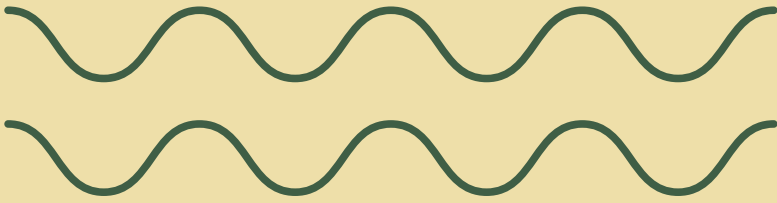
By clicking on the "Close Quiz" button, the user is redirected back to the literacy hub page



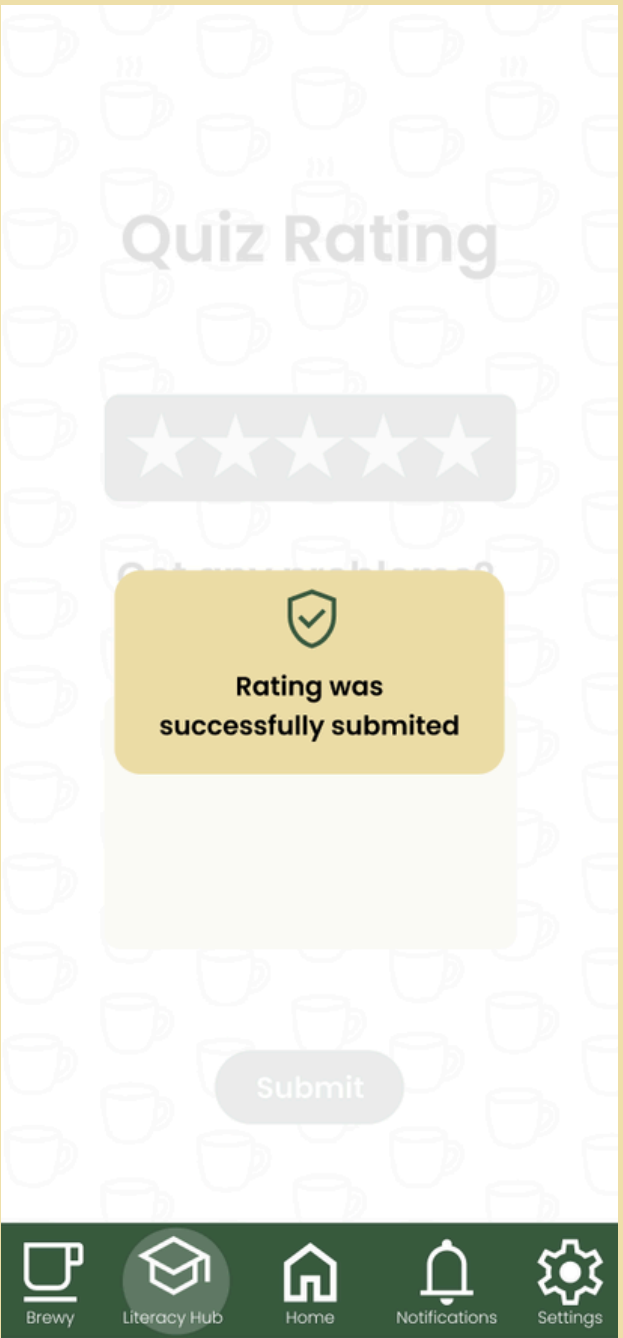
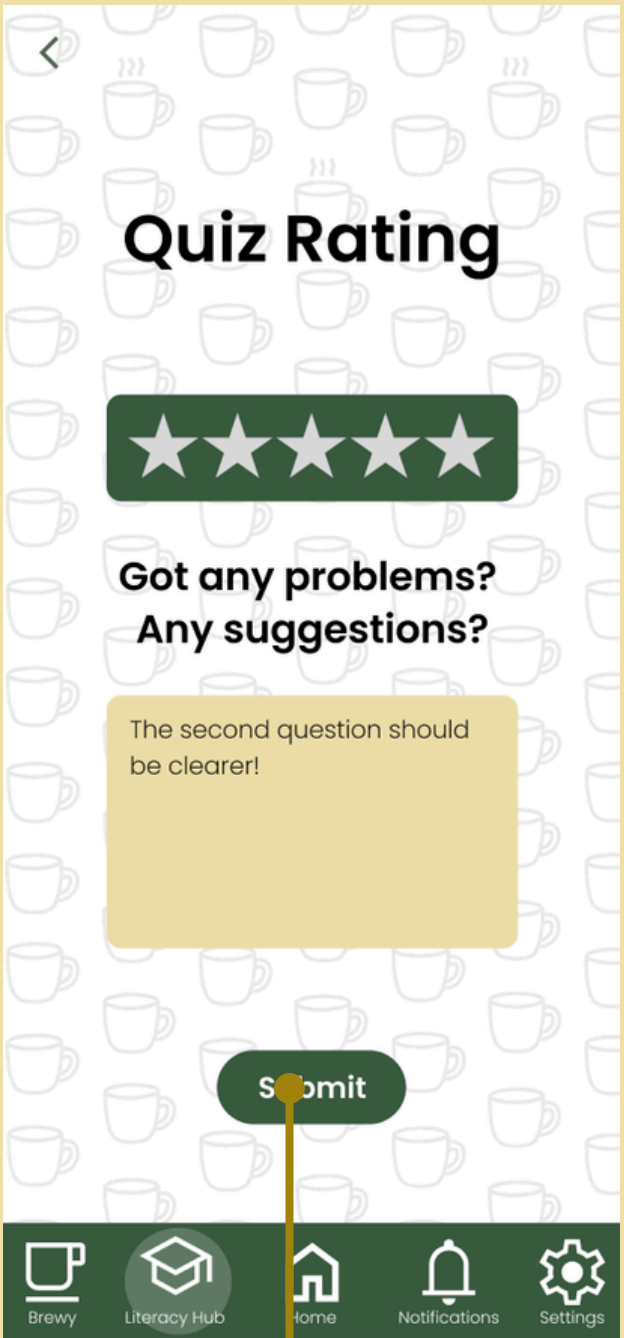
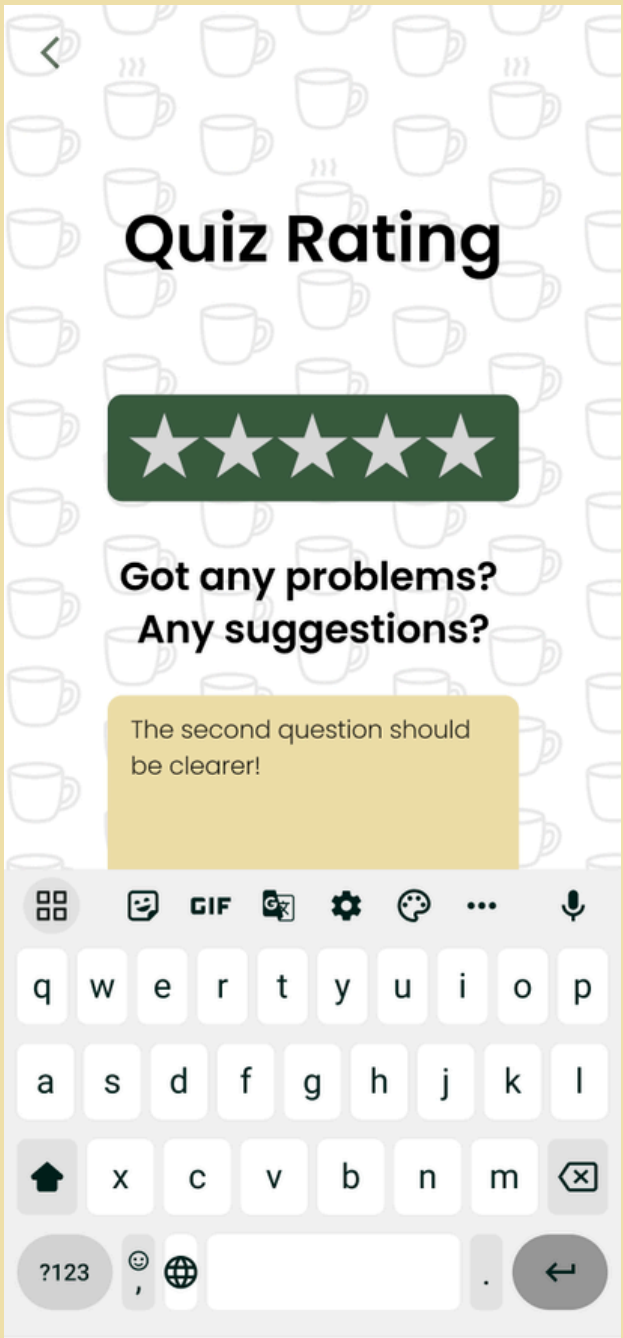
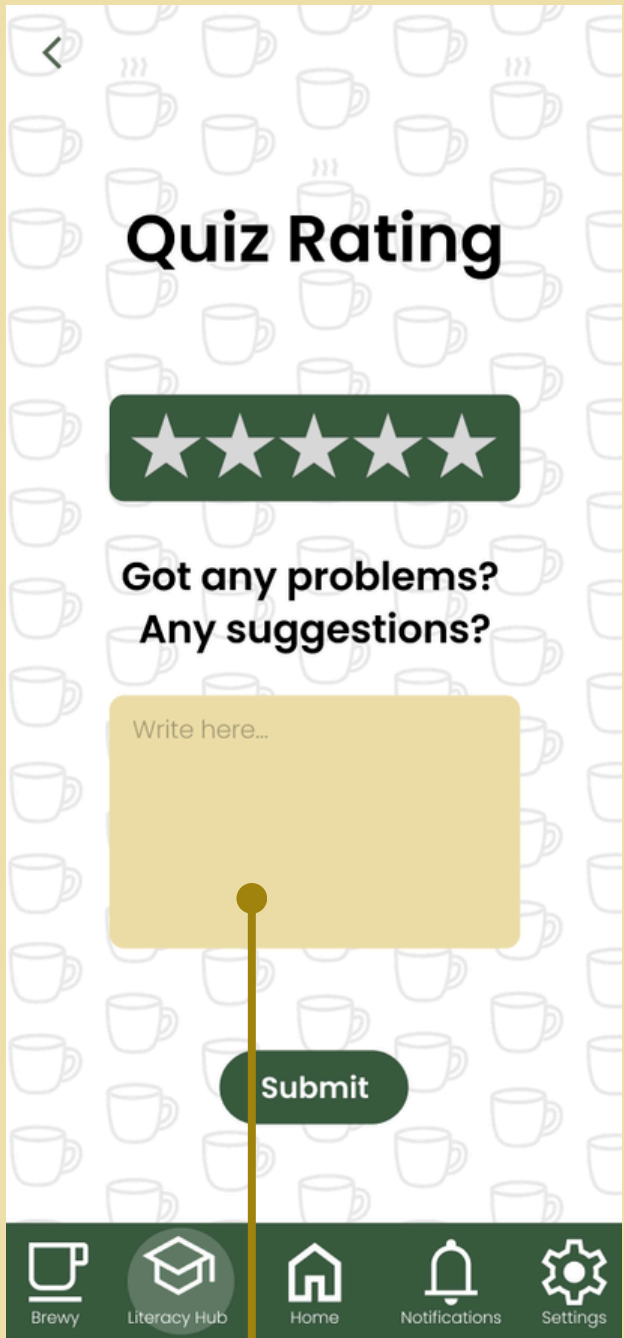
By clicking on the "Rate the quiz" button, the user is redirected to the rating page

*the correct option is highlighted for UI evaluation purposes

Wireflow - Financial Literacy Hub (cont.)



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Clicking on the feedback card opens a keyboard, which the user can use to write

By clicking on the “Submit” button, the user sees a confirmation page, then they are redirected to literacy hub initial page



Sample Characterization

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Sample size = 10 participants

Age Distribution 18-25 years: 30% (3 participants) 26-40 years: 40% (4 participants) 41+ years: 30% (3 participants)	Educational Background Basic education: 20% (2 participants) Secondary education: 50% (5 participants) Bachelor's degree: 30% (3 participants)
Employment Status Student: 10% (1 participant) Working student: 10% (1 participant) Employed: 80% (8 participants)	Mobile Technology Experience (scale 1-5) Score 2: 10% (1 participant) Score 3: 10% (1 participant) Score 4: 30% (3 participants) Score 5: 50% (5 participants)

Statistics

- Mode: 5
- Median: 4.5
- Q1 (25th percentile): 3.75
- Q3 (75th percentile): 5.0
- Mean: 4.3/5

Prior Experience with Financial Management Apps

Never used: 60% (6 participants)
Previously used: 40% (4 participants)

Demographic Summary:

The sample represents a **balanced, employed population with strong technological literacy** and diverse educational backgrounds. The mix of experienced and inexperienced app users allowed for comprehensive evaluation of interface intuitiveness across different user profiles. **Participants were representative of MoneyBrew's target audience:** busy, employed adults aged 18-40+ concerned with personal financial management.

Results & Statistical Analysis

TASK 1 (BREWY CHATBOT - 50-30-20 MODEL EXPLANATION)

Completion & Efficiency

Success rate: 90% (9/10 successfully completed)
Mean completion time: 22.79 seconds (SD:10.748)
Median: 19.00 seconds
Range: Q1 = 13.50s, Q3 = 33.10s
Unintentional clicks/misclicks: Only 1 across entire sample

Comparison with Established Requirements

Requirement	Target	Achieved	Status
Success rate	≥80%	90%	✓ Excellent
Completion time	≤2 minutes	22.79 seconds	✓ Well below
Average taps/clicks	≤5	1 misclick	✓ Excellent
Average rating	4-5/5	4.8/5	✓ Excellent

User Satisfaction (1-5)

Metric	Mode	Distribution	Mean
Efficiency	5	90% rated 5	4.9/5
Speed	5	50% rated 5, 30% rated 4	4.4/5
Ease	5	90% rated 5	4.9/5
Overall Experience	5	80% rated 5, 10% rated 4	4.8/5
Visual Appearance	4	60% rated 4, 20% rated 5	4.1/5

Key Findings

- **Task 1 exceeded all established requirements**
- Chatbot interface was highly intuitive with **minimal user errors**
- Response time from AI was perceived as **fast and satisfactory**
- Users found the task very **straightforward** with minimal cognitive load
- Visual design of chatbot interface received solid **positive feedback**

TASK 2: INACTIVE SUBSCRIPTION REMOVAL
(Smart Notifications)

TASK 3: FINANCIAL LITERACY HUB
(Work Contracts Content + Quiz)

Completion & Efficiency

- > **Success rate:** 100% (10/10)
 - > **Mean completion time:** 15.50 seconds (SD: 6.460)
 - > **Median:** 15.80 seconds
 - > **Range:** Q1 = 9.56s, Q3 = 19.03s
 - > **Misclicks:** 2 total (1 intentional, 1 irrelevant)
- > **Fully completed:** 80% (8/10)
 - > **Partially completed:** 20% (2/10)- completed quiz but skipped rating
 - > **Mean completion time:** 61.04 seconds (SD: 35.579)
 - > **Median:** 58.50 seconds
 - > **Range:** Q1 = 29.40s, Q3 = 88.12s
 - > **Misclicks:** 5 total (2 accidental, 1 intentional, 2 irrelevant)

User Satisfaction (means)

- > **Efficiency:** 4.9/5 | **Speed:** 5.0/5 | **Ease:** 4.7/5
 - > **Overall Experience:** 4.9/5 | **Visual Appearance:** 4.4/5
- > **Efficiency:** 4.6/5 | **Speed:** 4.6/5 | **Ease:** 4.8/5
 - > **Overall Experience:** 4.9/5 | **Visual Appearance:** 4.4/5

Status

✓ Exceeded all requirements
(100% success, <2 min completion, ≤5 clicks, 4-5/5 ratings)

✓ Met requirements
(80% full completion, within expected time range, positive ratings)

Key Insight: Smart notification feature was **highly effective**, since users immediately understood the recommendation and acted on it decisively.

Key Insight: Functionality works well, but the learning hub would benefit from **visual design enhancement** and **clearer navigation structure**.

Results & Statistical Analysis (cont.)

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Statistical Inference and Overall App Assessment

Statistical Analysis

All demographics (age, education, tech experience) showed NO significant differences ($p > 0.05$). Therefore, design is universally accessible & equally intuitive (caution due to small sample size)

User Adoption

Would Use App: 100%
Willingness: Perfect
Strong Product-Market Fit

Key Metrics

Success Rate: 90%
Visual Design: 4.1/5
Overall Experience: 4.3/5
Satisfaction: 4.3-4.8/5
Performance: Well below targets

